Online Appendices

Contents

A Paths to Power Codebook i
B List of variables in the individual-level dataset vii
C List of variables in the country-year dataset xiii
D Share missing by country and variable xvii
E Testing for patterns in missing data xx
F Extension of Besley and Reynal-Querol (2011) xxiv
G Inter-coder reliability tests xxvii

A Paths to Power Codebook

This codebook displays the questions, answer categories, and clarifications for all variables in the Paths to Power (PtP) dataset. For each country included, there is a list of cabinet ministers for the period 1966-2021, obtained from the WhoGov dataset. The list is rank-ordered from the "most important" person to the least important, where this rank, more specifically, is calculated using a score that considers both the importance of ministries and the number of years the person has been in cabinet.

Coders were asked to start with the most important person for a given country, Google the person, and then code the variables if there is sufficient information. The procedure is then repeated for the second-most important person on the list, and so on, until all registered ministers for the country are assessed.

If coders were unable to find sufficient information to code a variable for a particular minister, they were asked to write NA in the cell.

1. Education

1.1. What is the highest level of education completed by the cabinet minister? (Variable name: degree)

Clarification: Please code the highest level achieved prior to the person being employed as cabinet minister the first time. We follow the ISCED (2011) classification. ISCED level 0-1: Primary Education; level 2-3: Secondary Education; level 4-5: Post-secondary non-tertiary or short-cycle

tertiary education (typically practically-based, occupationally-specific educations); level 6: Bachelor's degree or equivalent; level 7: Master's degree or equivalent, this includes JD; level 8: Doctoral degree or equivalent. We have divided post-secondary non-tertiary into blue-collar educations (where manual labour is required, such as brick-layer, agricultural worker or gardener) and white-collar educations (desk, managerial, or administrative work, such as teacher, bank assistant or journalist. This group also includes ministers educated [outside of a university] to be nurses, police, and military officers).

Answer categories:

1. No or unfinished primary education
2. Primary education
3. Secondary education
4. Blue-collar post-secondary non-tertiary/short education
5. White-collar post-secondary non-tertiary/short education
6. Did not go to university, unclear which/any degree
7. Bachelor's degree
8. Master's degree
9. Doctoral degree
10. Went to university, unclear which/any degree

1.2. What type of degree was completed by the cabinet minister? (Variable name: degreetype)

Clarification: This question should only be coded for ministers scored 4-10 on question 1.1, and the question thus pertains to the highest level achieved prior to the person being employed as cabinet minister. If the minister has achieved multiple degrees at the same level, please code the last degree the minister obtained prior to becoming minister.

Answer categories:

1. Military
2. Mathematics/Computer science
3. Bio/Chem/Physics
4. Agronomy
5. Medicine
6. Engineering
7. Other natural science degree
8. Philosophy
9. Religion/theology

10. Other humanities degree
11. Law
12. Economics/Business/Management
13. Political science
14. Other social science degree
15. Other type of degree
16. Unclear content degree

1.3. What is the name of the university in which the minister received his/her degree? (Variable name: university)

Clarification: This question should only be coded for ministers coded on question 1.2, and the question pertains to the degree registered in that question. You may use (copy-paste) the name given in the source material, independent of language.

Open text answer

1.4. Did the minister study abroad? (Variable name: abroad)

Clarification: This question should only be coded for ministers scored 7-10 on question 1.1. The question should be answered yes if the person has studied abroad for at least one year at some period during his/her studies, or answered no if the person has not.

Answer categories:

1. No
2. Yes

1.5. In which country did the minister study abroad? (Variable name: abroadcountry)

Clarification: If the person has studied abroad, please insert the country name. You do not need to follow a particular way of writing the name of the country, since we will clean the variable at a later stage. If the person has studied abroad in multiple countries, then insert the country of the last degree. Please use the reported name of the country at the time of the minister's study there (example: "USSR").

Open text answer

2. Social background questions

2.1. Place of birth (Variable name: placeofbirth)

Clarification: If possible, please insert the city, town, village, or area where the cabinet minister was born or grew up. Please aim for the geographic level closest to the city-level (To give an example, John F. Kennedy is born on 83 Beals Street, Brookline, Massachusetts. Here Brookline, Massachusetts is sufficient). If there is no information on the city, please insert the region/state. If the person is born abroad, please note the country as well. If the city name has since changed, please use the modern name of the city (example: "Kristiania" should be coded "Oslo")

Open text answer

2.2. Is the cabinet minister a nobility or a member of the royal family? (Variable name: royal)

Clarification: A cabinet minister is a nobility/royalty if they are a part of the immediate or extended family of the king/queen, emir/emiras, sultans/sultanas, village/tribal kings. The question is not relevant in countries without a royal family.

Answer categories:

1. No
2. Yes

2.3. Does the person come from a family of politicians? (Variable name: politicalfamily))

Clarification: The person is coded as coming from a political family if at least one of the person's parents or grandparents held political office during their lifetime. Please note that caretakers being members of a political party, adherents to a particular ideology, or otherwise politically active – without holding office – is insufficient for coding "Yes". If the family was involved in politics in another country, we code it as a "Yes" if the other country is a pre-cursor state (for example Russia and the Soviet Union) or the former colonial power (for example France and Senegal).

Answer categories:

1. No
2. Yes

2.4. What is the cabinet minister's class background? (Variable name: classorigins)

Clarification: With "background" we are referring to the minister's upbringing from birth to his/her late teenage years. We want to evaluate the status of the family/care-taking unit where the minister spent most of his/her time during these years, regardless of whether this was a family with two or more caretakers (typically mother and father) or a single caretaker. If the class status of the caretakers differ (e.g., father is middle class, mother is lower class), please code the "highest" class. If the class status changes during the upbringing, please consider the status that pertains to the longest time interval (if time intervals are about evenly long, please code class status for the earliest time interval of the upbringing). Please note that if a minister's family has moved up or

down in class after the minister became an adult, we still code the class background at the time of the upbringing.

There are two dimensions to this question: The first pertains to the class background (correlated with, e.g., family income, education, property ownership, white vs blue-collar job). Thus, we primarily focus on economic class on this dimension. Insofar as economic class and other notions of class (e.g., networks with elites and part of "cultural upper class", but economically part of middle class), we code according to economic class.

The second dimension pertains to the sector in which the caretaking unit worked (primary sector vs. other sectors). The primary sector is here defined as farming (of different kinds), fishery, and forestry. Thus, this dimension mainly relates to whether the family worked with exploiting natural resources or not, and may also be highly correlated with geography (land or city). You may draw on the different correlates/clues in coding this question if class background and geography of upbringing is not explicitly mentioned in the available sources.

Examples of occupations regarded as typical primary sector lower class jobs: Peasants, farm workers, lumberjacks, fisher owning no or only small boat.

Example of occupation regarded as typical primary sector middle class jobs: Farmer (owns decent plot of land).

Examples of occupations regarded as typical primary sector upper class jobs: Plantation owner, largescale land/estate owner.

Examples of occupations regarded as typical lower class jobs in other sectors: Industrial worker, low-paid service jobs not requiring formal education (incl. in informal sector). Please note that mining (minerals, coal, petroleum) is also considered under this category (and not in the primary sector).

Example of occupation regarded as typical middle class jobs in other sectors: Academic, urban professionals (journalists, lawyers, doctor, etc.), small-scale business owner.

Examples of occupations regarded as typical upper class jobs in other sectors: CEO/owner large private company, Ambassador, Member of Parliament, Cabinet minister.

If sources do not mention sector or type of occupation at all, but just refers to "lower class/working class", "middle class", "upper class", etc., you should use categories 4-6. The exception is when the family is only referred to as "nobility" or "aristocracy" in the sources, in which case category 3 should be used (if these terms are used together with more specific occupations, then use the specific occupations as basis for coding).

Answer categories:

1. Lower class, primary sector
2. Middle class, primary sector
3. Upper class, primary sector
4. Lower class, other sectors
5. Middle class, other sectors
6. Upper class, other sectors

3. Occupation questions

3.1. What was the main occupation of the cabinet minister prior to becoming a full-time politician? (Variable name: occupation)

Clarification: With main occupation, we refer to the work-related activity the minister spent most of his/her time on in the decade before becoming a full-time politician. If a minister had two or more jobs the last decade and it is unclear from the sources which was the "main" one, please use either "Other blue collar job" or "Other white collar job", depending on the nature of the relevant jobs (if there is a mix of blue- and white collar jobs, choose the white collar category). We do not include previous political careers in this question, since they are left to question 3.2. If unable to distinguish between a 15 or 16, give preference to coding 15.

Answer categories:

1. Academic (research and tertiary education)
2. Student
3. Teacher in primary or secondary education
4. Priest or other religious occupation
5. Journalist
6. Judge or lawyer
7. Ambassador or working in the foreign service
8. Work in NGO or other non-profit organization
9. Work for a trade union
10. Work for a political party or an organization with strong ties to a party
11. Medical doctor or other health care worker
12. Military officer or soldier (army, navy and airforce)
13. Police or internal security (FBI, FSB, Stasi)
14. Blue-collar worker in industrial sector (incl. mining and construction)
15. CEO/CFO/VP/Head of Large Branch/Senior Management/Owner large private company (more than 50 employees)
16. Owner/manager small private company (incl. restaurant, café, shop) (less than 50 employees)
17. Chief or other informal/formal leadership role ethnic/geographically concentrated group
18. Large-scale agricultural land-owner/plantation owner/forest owner
19. Family farmer (owns decently sized plot of land)
20. Other agricultural worker (incl. peasants, farm workers, fishermen and lumberjacks)
21. Other white-collar job (private sector)
22. Other white-collar job (public sector)
23. Other white-collar job (unknown sector)
24. Other blue-collar job

25. No employment (incl. unemployed or stay-at-home mom or dad) or only holding a political office.
26. Job in popular culture (such as artist, television personality, or athlete)

3.2. Does the cabinet minister have political experience prior to being appointed minister? (Variable name: politician)

Clarification: With prior political experience, we refer to political activity taking place before the person was appointed to the cabinet. If the person both 1) is a partisan and 2) has held previous political office, having held political office trumps partisan.

Answer categories:

1. No prior political experience
2. Partisan without holding a political office (advisor, active in party organization, etc.)
3. Has previously held a political office (Member of legislature, local/regional government)

B List of variables in the individual-level dataset

B.1 Core variables

The PtP-dataset adds 11 new core variables to the set of original WhoGov-variables for each minister. We refer to (Nyrup and Bramwell 2020) for a description of the original variables included in WhoGov. The added core (PtP) variables are as follows:

degree A categorical variable with 10 categories that indicates the highest level of education completed by the cabinet minister. See Appendix A for an oversight of the categories.

degree_text The *degree*-variable with labels.

degreetype A categorical variable with 16 categories that indicates the type of degree completed by the minister, e.g. law or medicine. See Appendix A for an oversight categories.

degreetype_text The *degreetype*-variable with labels.

university A string variable which contains the name of the university (in English) where the minister received their degree. This variable has been cleaned so it follows the spelling used in the World Higher Education Database (WHED 2024).

abroad A binary variable which takes the value 1 if the minister studied abroad and 0 otherwise.

abroadcountry A string variable which contains the name of the country or countries where the minister studied abroad. The variable has not been thoroughly cleaned, and we refer to the derived variables (see below) for further information on this variable, such as the isocode for the given countries.

placeofbirth The city, town, village or area where the cabinet minister was born. We refer to the added variables (see below) for further information on this variable, such as geolocation.

royal A binary variable which takes on the value 1 if the minister is a nobility/royalty and 0 otherwise.

politicalfamily A binary variable which takes the value 1 if at least one of the person's parents or grandparents held political office during their lifetime and 0 otherwise.

classorigins A categorical variable with 6 categories that indicates the class background of the minister when growing up. The variable contains information both on the class background (Lower class, Middle class, Upper class) and the sector background (primary sector/other sectors). See Appendix A for an oversight of categories.

classorigins_text The *classorigins*-variable with labels.

occupation A categorical variable with 26 categories indicating the main occupation of the minister prior to becoming a full-time politician. See Appendix A for an oversight categories.

occupation_text The *occupation*-variable with labels.

politician A categorical variable with 3 categories indicating the political experience of the minister prior to joining the cabinet. The three categories are 1. No prior political experience, 2. Partisan without holding a political office (advisor, active in party organization, etc.), and 3. Has previously held a political office (Member of legislature, local/regional government).

politician_text The *politician*-variable with labels.

B.2 Derived variables

We have derived 20 additional variables using information from both PtP and WhoGov. These derived variables are the following:

highereducation A binary variable indicating whether a minister has *not* attended higher education (degree <=6) or has attended higher education (degree >=7).

degreetype_grouped A grouped degree variable with 6 categories. See Appendix B.3 for detailed information about the coding of this variable.

id_uni A unique university ID from the World Higher Education Database (WHED 2024), which can be used to merge Paths to Power with this dataset.

unitype A categorical variable with 7 categories indicating the type of university. By default, we code a university as university if it is found in the World Higher Education Database (WHED 2024). The categories are as follows:

1. University
2. University, not found in database
3. University, ceased existing
4. Military academy/military colleges/school/police academy
5. Party school
6. Specialized law school
7. Other
8. Universities/schools with same name, unknown which location
9. Unknown which university/school

abroad_isocode1 A variable containing the country's alpha-3 ISO code of the first country, where the person studied abroad.

abroad_isocode2 A variable containing the country's alpha-3 ISO code of the second country, where the person studied abroad.

abroad_isocode3 A variable containing the country's alpha-3 ISO code of the third country, where the person studied abroad.

abroad_isocode4 A variable containing the country's alpha-3 ISO code of the fourth country, where the person studied abroad.

abroad_isocode5 A variable containing the country's alpha-3 ISO code of the fifth country, where the person studied abroad.

pob_longitude The longitude for the person's place of birth.

pob_latitude The latitude for the person's place of birth.

pob_country The country where the person is born based on the place of birth.

pob_address_google The address provided by Google Maps for the place of birth which is used to calcualte the longitude and latitude.

pob_distancetocapital The place of birth's distance to the capital in kilometers. We use the primary capital according to the World Cities Database in 2024.

classorigins_grouped A nominal variable with three categories indicating the class background of the minister when growing up. The three categories are as follows:

1. 1. Lower
2. 2. Middle
3. 3. Upper

sector A binary variable indicating the sector background of the minister when growing up. The two levels are "1. Primary" and "2. Other".

occupation_6categories An occupation variable with 6 categories based on the coding in Gerring et al. (2019). See Appendix B.3 for detailed information about the coding of this variable.

occupation_10categories An occupation variable with 10 categories. See Appendix B.3 for detailed information about the coding of this variable.

classoccupation The occupational class of the minister. See Appendix B.3 for detailed information about the coding of this variable.

military A binary variable which takes the value 1 if the person 1) has a military title (var: title), has taken a military degree (degree_type = 1. Military), or has a military occupation (occupation = 12. Military officer or soldier (army, navy and airforce)), and otherwise 0.

Table B1: Construction of alternative degree type variable (degreetype_grouped)

Recoded degree type variable	Original variable
1. Military	1. Military
2. Natural Sciences	2. Mathematics/Computer science 3. Bio/Chem/Physics 4. Agronomy 5. Medicine 6. Engineering 7. Other natural science degree
3. Humanities	8. Philosophy 9. Religion/theology 10. Other humanities degree
4. Social Sciences	11. Law 12. Economics/Business/Management 13. Political science 14. Other social science degree
5. Other type of degree	15. Other type of degree
6. Unclear content degree	16. Unclear content degree

Table B2: Construction of alternative 6-group occupational variable (Occupation_6categories), inspired by (Gerring et al. 2019)

Recoded occupational background variable	Original occupation variable
1. White-collar	4. Priest/religious occupation 6. Judge or lawyer 7. Ambassador/foreign service 8. NGO/other non-profit org. 9. Trade union 11. Medical doctor or other health care worker 15. CEO/CFO/VP/etc 16. Owner/manager small private company 17. Chief or other leadership 18. Large-scale land-owner 21. Other white-collar job (private sector) 22. Other white-collar job (public sector) 23. Other white-collar job (unknown sector)
2. Blue-collar	14. Blue-collar worker industrial sector 19. Family farmer (owner) 20. Other agricultural worker 24. Other blue-collar job
3. Education	1. Academic (research/tertiary edu.) 2. Student 3. Teacher in primary or secondary education
4. Media and popular culture	5. Journalist 26. Job in popular culture (artist, TV, athlete)
5. Military and police	12. Military officer or soldier 13. Police
6. None or politics	10. Political party or org. 25. No employment

B.3 Further information on the derived variables

The tables below show the groupings (of original categories on core variables) that the derived variables are based on.

In the paper, we discussed and used tripartite class index based on the PtP occupational categorization and educational data. We note that this variable is about the minister's own occupational class, derived from their main occupation in the decade before entering politics, unlike the "class origins" variable in the dataset, which is based on the minister's upbringing/parental class. Table

Table B3: Construction of alternative 10-group occupational variable (occupation_10categories)

Recoded 10-group occupational background variable	Original occupation variable
1. Research and education	1. Academic (research/tertiary edu.) 2. Student 3. Teacher in primary or secondary education
2. Spiritual and ethnic/tribal leader	4. Priest/religious occupation 17. Chief or other leadership
3. High prestige white collar occupation	6. Judge or lawyer 11. Medical doctor or other health care worker 7. Ambassador/foreign service
4. CEO or landowner	15. CEO/CFO/VP/etc 18. Large-scale land-owner
5. Political organization	8. NGO/other non-profit org. 9. Trade union 10. Political party or org.
6. Military and police	12. Military officer or soldier 13. Police
7. Blue-collar occupation	14. Blue-collar worker industrial sector 19. Family farmer (owner) 20. Other agricultural worker 24. Other blue-collar job
8. White-collar occupation	16. Owner/manager small private company 21. Other white-collar job (private sector) 22. Other white-collar job (public sector) 23. Other white-collar job (unknown sector)
9. Media and Popular Culture	5. Journalist 26. Job in popular culture (artist, TV, athlete)
10. No prior employment	10. No employment

B4 provides a comprehensive and more precise description of how ministers are categorized into the different class categories. Most occupational groups can easily be classified into one of the three categories, upper- and upper-middle class, middle- and lower-middle class, or working class. For instance, placing CEOs or large-scale land owners in the upper- or upper-middle class category is a straightforward decision that arguably has high face validity. The same is the case, we surmise, for placing teachers in the middle- or lower-middle class category or for placing blue-collar workers in the working-class category.

However, some categories are more problematic to classify. There are essentially two groups of occupational categories that are hard to place. The first concerns aggregate occupational groups that incorporate occupations with different positions in society. One such example would be “medical doctors and other health care professionals", covering widely different groups such as medical doctors (who are upper- or upper-middle class) and unskilled nursing assistants (who are working class). In these cases, we rely on extra information from the education variables to categorize the individual ministers. For the medical sector, those with an advanced degree (MA or PhD) are classified as upper/upper-middle class, based on the assumption that most people with such degrees working in this sector will be in upper class occupations – most typically, doctors.

At the other end of the scale, those without any post-secondary education are classified as working class. This would include, for instance, unskilled and semi-skilled health workers. In between, in the middle/lower-middle class category, we have coded those with post-secondary or some/lower tertiary education, typical of, e.g., skilled nurses, as well as those with incomplete information about education (e.g., unclear degree, missing).

The “military officer or soldier" category is similarly heterogeneous class-wise as the discussed medical category. Also here, we divide the broader category by the ministers’ education, but now

so that individuals with any higher education are considered upper- or upper-middle class. Most of these will be officers, educated at military schools, but also with other relevant educations, for instance within law or engineering.

Those with no education or only primary or secondary education are considered to be working class, assuming that they are soldiers. Finally, those with post-secondary, non-tertiary education are considered lower-middle/middle class along with those with incomplete information about education (for instance, without higher education but uncertain which degrees they have). These assumptions are, of course, not always true, as one could work one's way up to a high-level officer position without a higher education in some military systems. However, we consider the likely error from following this assumption to be smaller than if we were to lump all soldiers and officers together in the same class. Yet scholars using this index for the purpose of capturing, e.g., the class basis of governments in military dictatorships, where these groups are likely to make up a larger share of the ministers, may want to be extra careful and check whether codings hold based on more in-depth knowledge of the country and ministers in question.

A second group of hard-to-classify cases is the more abstract categories pertaining to work for a particular type of entity or in a sector that contains a wide variety of positions. For instance, working in an NGO, a political party, or police/internal security could imply anything from being a low-level clerk or police officer to being a leader of a large NGO or the head of the FSB in Russia. Clearly, some of these are upper/upper-middle class jobs while others are middle/lower-class jobs. Since there is no obvious way, with the data we have at hand, to identify which group these ministers belong to, we place them all in the middle category. Alternative assumptions, and thus versions of our schema, could easily be made. For the latter example, we could, e.g., code all such ministers as upper/upper-middle class or we could use education levels to separate ministers belonging to these heterogeneous categories. Table B4 shows the decision-rules behing ministers' being placed in the different categories.

Finally, the "other white-collar jobs" (whether in private, public, or unknown sector) categories are related to similar problems as those discussed above, but come with additional important trade-offs. These categories are coded as middle- or lower-middle class in our current schema. However, especially in modern economies, some of these occupations are functionally equivalent to the working class (e.g. service workers in Oesch's schema and lower-grade routine non-manual workers, that is class IIIb, in the EGPs chema). This means that the first version of our schema creates a somewhat "traditional" working-class category with blue-collar and farm labour occupations being counted, but excluding some (often more female-dominated) service-worker occupations that are prevalent in modern societies.

The advantage of this choice is that most individuals who are coded as working class are, indeed, working class following most conventional occupation-based class schemes, and this holds true across temporal and geographic contexts. The disadvantage is that it excludes some people that should – particularly in the modern era and in richer countries – be considered working class, notably service workers.

We therefore add an alternative measure in the dataset where we code people in a similar fashion as for those in the medical sector: Those in white-collar jobs with low (no, primary, or secondary) education are coded as working class; while those with advanced degrees (MA or PhD) are coded as upper or upper middle class. The rest remain in the middle/lower-middle category. The advantages and disadvantages of this approach mirror those of our first approach: The advantage is that it is inclusive also of groups that would be considered working class in some time periods and some

geographical areas. The disadvantage is that it also includes some groups that are functionally more akin to the middle class as working class. The errors related to this trade-off may vary by country and over time. In a poor country where few people have higher education, an office clerk with secondary education would arguably be considered a middle-class individual, whether one considers marketable skills (as in the Oesch schema) or employment contract type (as in the EGP schema) as the defining vertical criterion. The same was probably true for many currently wealthy countries, e.g. in Europe, in the 1960s and early 1970s.

Following Adcock and Collier's (2001) influential work on measurement validity, content validation is a matter of two questions: First, are key elements omitted from the indicator? Second, are inappropriate elements included in the indicator? The choice of working-class cut-off point is essentially a trade-off between these two: Our first schema does omit some relevant elements, but does not include many inappropriate elements (when the point of departure is correctly identifying working-class individuals). In contrast, the more inclusive variable omits fewer relevant elements, but includes more inappropriate elements. Our first version is probably more problematic for richer countries and more recent time periods, whereas the second version is probably more problematic for poorer countries and more distant time periods.

As mentioned in the main article, readers should be aware that this derived class measure is not measured without error, since it is built from an occupational categorization with some fairly broad categories. For instance, if we are interested in working-class representation in politics, the schema probably under-estimates the share of people with an occupational background from the working class. This is because the occupational variable codes the main occupation in the last ten years before entering politics. But some ministers started out in a working-class occupation in their youth, before moving on to white-collar jobs in, e.g., trade unions where they remained for years before entering politics – these individuals are thus not coded as working class, even though one may make the argument that they should have been (probably depending on the relative length of their careers in the different occupations).

Still, it does provide a relatively nuanced and comparable measure of occupational class that spans decades and continents, thus allowing for many new research questions to be answered. What is more, we highlight again that this particular class coding is just one example of how to re-code variables of theoretical interest from PtP's occupation or other variables. The flexibility of PtP's many multi-category variables allows researchers to create derived variables according to different research needs. For example, if class is a key variable, one may want to code different categories of occupations into different classes for different time periods or geographical areas to take into account context-specific differences in class structures. Thus, for instance, people with white-collar jobs and no education could be considered working class in, say, European countries from the 1990s onwards, but middle class in poorer countries in the 1960s.

C List of variables in the country-year dataset

The individual-level dataset (i.e., with ministers as units of analysis) can be used to create a country-year dataset with information for the whole cabinet for every country. We have created a relatively limited set of country-year variables but users can easily create additional variables. First, we have a group of variables that are general (General variables), such as missingness values for individual country-years and information about the leader. Next, we have created a list of variables for three different groups of government members based on the definitions in WhoGov. These are 1) all people in the dataset (total), 2) core members (core), and 3) ministers. We only list the information

Table B4: Construction of the occupational class variable (classoccupations)

Recoded class variable	Original occupation variable	Comments
1. Upper class/upper middle class	1. Academic (research/tertiary edu.)	-
	6. Judge or lawyer	-
	7. Ambassador/foreign service	-
	11. Medical doctor or other health care worker	If MA/PhD degree
	12. Military officer or soldier	If any higher edu.
	15. CEO/CFO/VP/etc	-
	17. Chief or other leadership	-
	18. Large-scale land-owner	-
2. Middle class/lower middle class	3. Teacher in primary/secondary edu.	-
	4. Priest/religious occupation	-
	5. Journalist	-
	8. NGO/other non-profit org.	-
	9. Trade union	-
	10. Political party or org.	-
	11. Medical doctor or other health care worker	If BA degree, post-secondary edu., or incomplete info. about edu.
	12. Military officer or soldier	If post-secondary non-tertiary edu. or incomplete info. about edu.
	13. Police or internal security	-
	16. Owner/manager small private company	-
	19. Family farmer (owner)	-
	21. Other white-collar job (private sector)	-
	22. Other white-collar job (public sector)	-
	23. Other white-collar job (unknown sector)	-
	26. Job in popular culture (artist, TV, athlete)	-
3. Working class	12. Military officer or soldier	If no, primary, or secondary edu.
	14. Blue-collar worker industrial worker	-
	11. Medical doctor or other health care worker	If no, primary, or secondary edu.
	20. Other agricultural worker	-
	24. Other blue-collar job	-
4. Uncodeable/no class	25. No employment or only pol. office	-
	2. Student	With no occupation, we cannot code class

for 1) here, but the same variables exist for ministers and core members. NAs are not included when calculating the share in each group. The variables are as follows:

C.1 General variables

year Year. We are using July for all years apart from 1966, where data was only available for September in the original WhoGov dataset, and 1970, where we are using January instead of July.

country_isocode The country's alpha-3 ISO code.

coder_id Unique ID for each coder.

coderdifficulty The coder's qualitative assessment of the difficulty of coding the country with the scores 1 (easy), 2 (medium), and 3 (difficult).

n_total Number of individuals in the dataset, by country and year

degree_missing Share missing for *degree*, by country and year.

degreetype_missing Share missing (out of those who went to university) for *degreetype*, by country and year.

university_missing Share missing (out of those who went to university) for *university*, by country and year.

abroad_missing Share missing (out of those who went to university) for *abroad*, by country and year.

abroadcountry_missing Share missing (out of those studying abroad) for *abroadcountry*, by country and year.

placeofbirth_missing Share missing for *placeofbirth*, by country and year.

royal_missing Share missing for *royal*, by country and year.

politicalfamily_missing Share missing for *politicalfamily*, by country and year.

classorigins_missing Share missing for *classorigins*, by country and year.

occupation_missing Share missing for *occupation*, by country and year.

politician_missing Share missing for *politician*, by country and year.

leader_name The leader's name.

leader_gender The leader's gender

leader_born The leader's year of birth.

leader_dead The leader's year of death.

leader_party The leader's political party.

leader_degree The leader's degree.

leader_degreetype The leader's degree type.

leader_uni The leader's place of studying.

leader_abroad Whether the leader studied abroad.

leader_abroadcountry Where the leader studied abroad.

leader_birthplace The leader's place of birth.

leader_royal Whether the leader is from a royal family.

leader_politicalfamily Whether the leader is from a political family.

leader_classorigins The leader's class background.

leader_occupation The leader's occupation prior to joining politics.

leader_politican The leader's political experience before joining the government.

C.2 Variables calculated for total, minister, and core members

nodegree_total Share of total cabinet with no education beyond secondary, by country and year

training_total Share of total cabinet with vocational training but no university degree, by country and year

degree_total Share of total cabinet with a university degree, by country and year

doctor_total Share of total cabinet with a doctorate, by country and year

militarydegree_total Share of total cabinet with a military education, by country and year

naturalsciencedegree_total Share of total cabinet with a natural science degree, by country and year

humanitiesdegree_total Share of total cabinet with a degree in humanities, by country and year

socialsciencedegree_total Share of total cabinet with a social science degree, by country and year

otherdegree_total Share of total cabinet with a degree in a different field than the ones mentioned, by country and year

unknowndegree_total Share of total cabinet that has a university degree of unknown type, by country and year

abroad_total Share of total cabinet that studied abroad for at least one year, by country and year

royal_total Share of total cabinet that comes from a royal family, by country and year

polfamily_total Share of total cabinet that comes from a political family, by country and year

workingclass_total Share of total cabinet that comes from a working-class background, by country and year

middleclass_total Share of total cabinet that comes from a middle-class background, by country and year

upperclass_total Share of total cabinet that comes from an upper-class background, by country and year

primary_total Share of total cabinet with a class background in the primary sector, by country and year

academic_total Share of total cabinet that previously worked in 1. Research and education, by country and year

spiritual_total Share of total cabinet that previously worked in 2. Spiritual and ethnic/tribal leader, by country and year

prestigiouswhitecollar_total Share of total cabinet that previously worked in 3. High prestige white collar occupation, by country and year

ceolandowner_total Share of total cabinet that previously worked in 4. CEO or landowner, by country and year

politicalorganisation_total Share of total cabinet that previously worked in 5. Political organization, such as working for a political party, by country and year

militarypolice_total Share of total cabinet that previously worked in 6. Military and police, by country and year

bluecollar_total Share of total cabinet that previously worked in 7. Blue-collar occupation, by country and year

whitecollar_total Share of total cabinet that previously worked in 8. White-collar occupation, by country and year

mediapopularcolture_total Share of total cabinet that previously worked in 9. Media and popular culture, by country and year

noemployment_total Share of total cabinet that previously worked in 10. No prior employment, by country and year

nopol_total Share of total cabinet with no prior political experience, by country and year

party_total Share of total cabinet that previously had been members of a political party, by country and year

pol_total Share of total cabinet that had held another political office prior to becoming a minister, by country and year

military_total Share of total cabinet that has a military background, by country and year

D Share missing by country and variable

In Table D1, we show the share of missing data by country for the 11 core variables. The missingness shares are calculated as the number of person-years with missing values divided by all person-years. Thus, a person who appears multiple times across years will receive more weight than a person who only held a ministerial post for a year. We exclude people who have not attended university when calculating missingness for *type of degree* and *university*. Moreover, we exclude people who did not attend university or did not study abroad when calculating missingness for *country of studying abroad*.

Table D1: Share missing

Country Isocode	Degree	Type of degree	University	Studied abroad	Country of studying abroad	Place of birth	Royal	From a political family	Class background	Previous occupation	Background in politics
AFG	0.22	0.25	0.40	0.34	0.34	0.34	0.19	0.25	0.80	0.22	0.25
AGO	0.16	0.22	0.56	0.29	0.29	0.20	0.00	0.20	0.71	0.17	0.19
ALB	0.14	0.22	0.30	0.26	0.25	0.16	0.00	0.10	0.70	0.13	0.13
ARE	0.19	0.32	0.40	0.36	0.36	0.25	0.05	0.15	0.34	0.12	0.14
ARG	0.09	0.09	0.16	0.34	0.34	0.11	0.00	0.65	0.56	0.11	0.26
ARM	0.18	0.18	0.19	0.19	0.19	0.19	0.00	0.82	0.89	0.24	0.27
AUS	0.07	0.07	0.08	0.08	0.07	0.02	0.01	0.27	0.28	0.03	0.02
AUT	0.00	0.13	0.13	0.15	0.15	0.00	0.00	0.27	0.27	0.00	0.00
AZE	0.04	0.04	0.05	0.04	0.04	0.06	0.00	0.04	0.38	0.04	0.04
BDI	0.49	0.50	0.65	0.64	0.64	0.67	0.00	0.54	0.91	0.45	0.56
BEL	0.00	0.04	0.09	0.03	0.03	0.00	0.00	0.00	0.00	0.00	0.00
BEN	0.35	0.35	0.56	0.54	0.54	0.52	0.77	0.78	0.95	0.39	0.40
BFA	0.30	0.31	0.51	0.48	0.48	0.57	0.20	0.37	0.90	0.30	0.34
BGD	0.25	0.26	0.33	0.34	0.34	0.14	0.13	0.49	0.59	0.24	0.11
BGR	0.22	0.23	0.27	0.24	0.23	0.36	0.26	0.26	0.55	0.19	0.22
BHR	0.13	0.21	0.30	0.26	0.26	0.48	0.05	0.07	0.43	0.14	0.14
BLR	0.07	0.09	0.10	0.10	0.10	0.08	0.01	0.77	0.77	0.06	0.07
BOL	0.36	0.43	0.57	0.60	0.60	0.49	0.27	0.81	0.82	0.38	0.65
BRA	0.01	0.01	0.03	0.01	0.00	0.01	0.02	0.05	0.05	0.01	0.01

Table D1: Share missing *(continued)*

Country Isocode	Degree	Type of degree	University	Studied abroad	Country of studying abroad	Place of birth	Royal	From a political family	Class back-ground	Previous occupa-tion	Background in politics
BTN	0.33	0.33	0.48	0.34	0.33	0.66	0.38	0.39	0.74	0.43	0.39
BWA	0.19	0.23	0.32	0.25	0.25	0.27	0.11	0.09	0.10	0.11	0.06
CAN	0.13	0.14	0.17	0.09	0.09	0.00	0.00	0.00	0.58	0.09	0.01
CHE	0.00	0.11	0.11	0.08	0.08	0.00	0.00	0.15	0.15	0.00	0.00
CHL	0.08	0.09	0.09	0.20	0.20	0.12	0.00	0.70	0.50	0.09	0.33
CHN	0.15	0.16	0.19	0.17	0.17	0.12	0.13	0.56	0.55	0.12	0.12
CIV	0.28	0.29	0.41	0.38	0.39	0.33	0.00	0.21	0.83	0.20	0.22
CMR	0.12	0.13	0.26	0.21	0.21	0.11	0.11	0.90	0.90	0.11	0.11
COD	0.40	0.41	0.56	0.54	0.53	0.49	0.00	0.35	0.88	0.36	0.33
COL	0.09	0.07	0.10	0.09	0.10	0.09	0.00	0.49	0.47	0.07	0.08
CRI	0.20	0.20	0.28	0.28	0.28	0.44	0.10	0.36	0.64	0.22	0.22
CUB	0.23	0.23	0.37	0.32	0.31	0.36	0.19	0.19	0.58	0.22	0.22
CYP	0.01	0.02	0.08	0.04	0.04	0.01	0.00	0.00	0.57	0.00	0.00
CZE	0.03	0.04	0.12	0.12	0.10	0.04	0.00	0.04	0.62	0.03	0.01
DDR	0.17	0.17	0.23	0.23	0.23	0.11	0.00	0.36	0.36	0.16	0.13
DEU	0.00	0.08	0.10	0.10	0.10	0.00	0.00	0.03	0.03	0.00	0.00
DNK	0.00	0.06	0.09	0.10	0.09	0.00	0.00	0.00	0.00	0.00	0.00
DOM	0.18	0.18	0.38	0.32	0.32	0.24	0.11	0.34	0.55	0.29	0.26
DZA	0.21	0.23	0.44	0.35	0.36	0.16	0.00	0.13	0.84	0.14	0.14
ECU	0.23	0.23	0.32	0.33	0.32	0.38	0.14	0.34	0.75	0.27	0.30
EGY	0.19	0.20	0.31	0.28	0.27	0.49	0.03	0.18	0.89	0.22	0.22
ERI	0.46	0.75	0.59	0.60	0.60	0.46	0.73	0.84	0.85	0.56	0.54
ESP	0.01	0.07	0.07	0.06	0.06	0.00	0.00	0.40	0.43	0.00	0.00
EST	0.01	0.01	0.01	0.00	0.00	0.00	0.00	0.00	0.59	0.01	0.00
ETH	0.41	0.46	0.46	0.44	0.45	0.46	0.62	0.77	0.78	0.43	0.43
FIN	0.00	0.01	0.14	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
FRA	0.06	0.11	0.27	0.21	0.21	0.01	0.00	0.20	0.32	0.03	0.01
GBR	0.00	0.00	0.07	0.01	0.01	0.01	0.08	0.08	0.08	0.01	0.00
GEO	0.03	0.03	0.04	0.04	0.02	0.06	0.00	0.03	0.70	0.03	0.05
GHA	0.29	0.28	0.33	0.32	0.32	0.31	0.16	0.72	0.77	0.29	0.24
GIN	0.45	0.46	0.64	0.61	0.60	0.58	0.87	0.85	0.89	0.48	0.54
GRC	0.06	0.06	0.11	0.04	0.05	0.07	0.00	0.04	0.29	0.02	0.02
GTM	0.30	0.39	0.56	0.60	0.61	0.69	0.21	0.83	0.85	0.34	0.62
HND	0.07	0.07	0.16	0.14	0.14	0.26	0.29	0.29	0.29	0.09	0.08
HRV	0.02	0.02	0.03	0.03	0.02	0.02	0.00	0.03	0.62	0.03	0.03
HTI	0.41	0.42	0.71	0.68	0.68	0.69	0.00	0.52	0.85	0.36	0.47
HUN	0.00	0.03	0.09	0.06	0.03	0.01	0.00	0.00	0.29	0.00	0.00
IDN	0.01	0.01	0.04	0.01	0.02	0.01	0.07	0.08	0.08	0.01	0.01
IND	0.09	0.31	0.14	0.19	0.19	0.07	0.11	0.55	0.52	0.25	0.07
IRL	0.00	0.00	0.03	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
IRN	0.06	0.06	0.17	0.10	0.11	0.06	0.04	0.05	0.67	0.06	0.06
IRQ	0.11	0.18	0.41	0.27	0.27	0.19	0.08	0.10	0.66	0.12	0.12
ISL	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ISR	0.04	0.09	0.16	0.16	0.16	0.03	0.00	0.35	0.40	0.04	0.03
ITA	0.02	0.05	0.17	0.05	0.05	0.02	0.02	0.43	0.43	0.02	0.02
JAM	0.18	0.17	0.28	0.26	0.26	0.24	0.00	0.25	0.55	0.17	0.17
JOR	0.10	0.11	0.21	0.15	0.15	0.31	0.10	0.16	0.78	0.10	0.12
JPN	0.00	0.00	0.01	0.00	0.00	0.03	0.00	0.00	0.28	0.04	0.00
KAZ	0.02	0.03	0.04	0.03	0.01	0.03	0.00	0.02	0.50	0.02	0.02
KEN	0.12	0.14	0.15	0.15	0.13	0.25	0.01	0.38	0.43	0.14	0.12
KHM	0.41	0.46	0.58	0.54	0.54	0.52	0.00	0.38	0.69	0.44	0.37
KOR	0.15	0.15	0.16	0.17	0.17	0.15	0.16	0.59	0.86	0.14	0.15
KWT	0.07	0.33	0.49	0.41	0.41	0.13	0.03	0.04	0.50	0.08	0.04
LBN	0.11	0.15	0.25	0.21	0.21	0.17	0.00	0.08	0.63	0.08	0.09
LBR	0.59	0.59	0.61	0.58	0.58	0.63	0.56	0.61	0.64	0.59	0.56
LBY	0.31	0.36	0.55	0.51	0.51	0.51	0.01	0.31	0.87	0.28	0.31
LKA	0.19	0.19	0.23	0.19	0.19	0.42	0.00	0.11	0.49	0.18	0.03

Table D1: Share missing *(continued)*

Country Isocode	Degree	Type of degree	University	Studied abroad	Country of studying abroad	Place of birth	Royal	From a political family	Class back-ground	Previous occupa-tion	Background in politics
LTU	0.01	0.01	0.01	0.01	0.00	0.01	0.00	0.01	0.63	0.01	0.01
LUX	0.06	0.12	0.18	0.17	0.17	0.02	0.00	0.64	0.65	0.01	0.01
LVA	0.07	0.07	0.08	0.07	0.04	0.13	0.00	0.08	0.42	0.08	0.10
MAR	0.18	0.19	0.35	0.28	0.27	0.23	0.15	0.15	0.86	0.15	0.17
MDA	0.04	0.04	0.05	0.04	0.03	0.15	0.02	0.04	0.13	0.04	0.05
MDG	0.37	0.39	0.55	0.54	0.54	0.63	0.00	0.41	0.85	0.31	0.48
MEX	0.03	0.04	0.08	0.34	0.34	0.04	0.01	0.67	0.63	0.09	0.41
MKD	0.03	0.03	0.07	0.07	0.05	0.07	0.00	0.03	0.78	0.06	0.08
MLI	0.26	0.26	0.38	0.33	0.33	0.36	0.00	0.96	0.95	0.28	0.28
MLT	0.06	0.07	0.13	0.12	0.12	0.13	0.00	0.04	0.67	0.05	0.00
MMR	0.66	0.75	0.66	0.67	0.67	0.62	0.69	0.70	0.94	0.49	0.53
MNE	0.23	0.23	0.30	0.30	0.28	0.27	0.07	0.89	0.93	0.33	0.37
MOZ	0.32	0.36	0.46	0.39	0.39	0.32	0.23	0.24	0.91	0.29	0.29
MWI	0.35	0.42	0.49	0.45	0.45	0.41	0.71	0.71	0.75	0.31	0.26
MYS	0.13	0.13	0.16	0.12	0.12	0.11	0.01	0.11	0.52	0.14	0.03
NAM	0.01	0.12	0.18	0.13	0.13	0.02	0.00	0.01	0.86	0.01	0.01
NER	0.43	0.44	0.60	0.55	0.55	0.53	0.47	0.92	0.93	0.42	0.44
NGA	0.18	0.18	0.21	0.20	0.20	0.22	0.24	0.24	0.64	0.16	0.17
NLD	0.02	0.11	0.13	0.14	0.12	0.01	0.01	0.45	0.49	0.01	0.01
NOR	0.00	0.01	0.06	0.05	0.02	0.00	0.00	0.00	0.00	0.00	0.00
NPL	0.48	0.48	0.56	0.55	0.55	0.22	0.13	0.69	0.81	0.53	0.26
NZL	0.05	0.05	0.08	0.06	0.07	0.03	0.27	0.27	0.32	0.02	0.00
OMN	0.31	0.51	0.62	0.56	0.56	0.45	0.10	0.18	0.36	0.22	0.26
PAK	0.28	0.28	0.36	0.35	0.35	0.25	0.22	0.42	0.46	0.36	0.21
PER	0.20	0.21	0.25	0.28	0.27	0.26	0.16	0.72	0.75	0.18	0.40
PHL	0.12	0.12	0.14	0.10	0.10	0.22	0.09	0.32	0.32	0.14	0.23
PNG	0.43	0.49	0.55	0.53	0.53	0.56	0.00	0.83	0.85	0.39	0.27
POL	0.02	0.03	0.04	0.03	0.03	0.02	0.19	0.19	0.56	0.02	0.02
PRK	0.54	0.57	0.62	0.59	0.58	0.55	0.03	0.85	0.85	0.50	0.55
PRT	0.01	0.07	0.10	0.08	0.07	0.05	0.04	0.46	0.54	0.02	0.02
PRY	0.21	0.23	0.47	0.53	0.52	0.40	0.00	0.56	0.80	0.30	0.35
QAT	0.18	0.36	0.43	0.40	0.40	0.23	0.08	0.15	0.27	0.22	0.18
ROU	0.02	0.02	0.03	0.02	0.01	0.10	0.02	0.02	0.11	0.03	0.03
RUS	0.17	0.17	0.18	0.21	0.19	0.15	0.00	0.41	0.41	0.15	0.15
RWA	0.38	0.37	0.44	0.42	0.42	0.53	0.84	0.84	0.81	0.35	0.39
SAU	0.05	0.26	0.27	0.07	0.07	0.13	0.04	0.06	0.09	0.04	0.06
SDN	0.26	0.27	0.44	0.40	0.40	0.50	0.25	0.31	0.86	0.28	0.34
SEN	0.19	0.19	0.36	0.34	0.34	0.25	0.22	0.82	0.82	0.17	0.16
SGP	0.02	0.03	0.02	0.02	0.01	0.02	0.00	0.08	0.08	0.03	0.02
SOM	0.72	0.73	0.74	0.73	0.71	0.72	0.00	0.71	1.00	0.71	0.71
SRB	0.12	0.14	0.01	0.14	0.13	0.01	0.61	0.62	0.78	0.15	0.13
SSD	0.32	0.33	0.48	0.41	0.41	0.46	0.04	0.26	0.90	0.24	0.19
SUN	0.36	0.36	0.38	0.40	0.40	0.35	0.00	0.53	0.53	0.38	0.35
SVK	0.02	0.02	0.02	0.02	0.01	0.03	0.01	0.01	0.43	0.02	0.06
SVN	0.03	0.03	0.07	0.05	0.05	0.11	0.00	0.01	0.59	0.02	0.11
SWE	0.12	0.13	0.20	0.20	0.19	0.05	0.00	0.34	0.34	0.07	0.03
SYR	0.30	0.32	0.48	0.39	0.39	0.42	0.11	0.28	0.78	0.31	0.32
TCD	0.48	0.55	0.68	0.61	0.62	0.59	0.34	0.46	0.89	0.39	0.41
THA	0.05	0.05	0.10	0.07	0.08	0.18	0.02	0.27	0.52	0.16	0.03
TJK	0.16	0.16	0.20	0.18	0.16	0.19	0.00	0.15	0.87	0.14	0.16
TUN	0.12	0.12	0.29	0.22	0.22	0.16	0.03	0.09	0.79	0.10	0.10
TUR	0.03	0.03	0.05	0.05	0.05	0.05	0.61	0.78	0.78	0.03	0.05
TWN	0.12	0.13	0.14	0.15	0.15	0.24	0.24	0.62	0.63	0.13	0.15
TZA	0.21	0.21	0.24	0.22	0.21	0.27	0.69	0.69	0.75	0.18	0.19
UGA	0.19	0.19	0.24	0.22	0.22	0.23	0.71	0.72	0.71	0.17	0.19
UKR	0.04	0.07	0.08	0.07	0.05	0.04	0.03	0.48	0.47	0.04	0.04
URY	0.12	0.15	0.29	0.30	0.30	0.20	0.00	0.35	0.67	0.25	0.27

Table D1: Share missing *(continued)*

Country Isocode	Degree	Type of degree	University	Studied abroad	Country of studying abroad	Place of birth	Royal	From a political family	Class back-ground	Previous occupa-tion	Background in politics
USA	0.00	0.01	0.01	0.01	0.01	0.01	0.00	0.01	0.06	0.00	0.02
UZB	0.25	0.25	0.28	0.25	0.24	0.26	0.00	0.18	0.60	0.17	0.18
VEN	0.25	0.28	0.38	0.44	0.43	0.45	0.23	0.79	0.79	0.27	0.53
VNM	0.12	0.23	0.44	0.36	0.36	0.03	0.00	0.05	0.64	0.07	0.03
YEM	0.24	0.28	0.53	0.45	0.46	0.29	0.18	0.22	0.89	0.23	0.27
ZAF	0.07	0.09	0.10	0.07	0.06	0.09	0.05	0.05	0.10	0.06	0.05
ZMB	0.35	0.36	0.48	0.49	0.49	0.46	0.70	0.73	0.74	0.27	0.26
ZWE	0.13	0.18	0.22	0.20	0.19	0.20	0.53	0.53	0.60	0.10	0.06
Average	0.17	0.20	0.27	0.25	0.25	0.23	0.13	0.34	0.58	0.17	0.18

E Testing for patterns in missing data

In this appendix, we assess patterns of missingness. We test for country-, time-, and individual-level factors that may predict missingness. Table E1 presents the results for the country-level analysis using country-year data, while Table E2 presents the results for the individual-level analysis. In both analyses we assess missingness for three main variables, namely degree, class origins, and occupation.

We find better data availability for more democratic countries and for richer countries, while there is no association between population size and missingness. Moreover, missingness is lowest in Europe and highest in Africa.

For the individual-level analysis, we find that data availability is better for leaders than for non-leaders, and for high prestige positions compared to both low- and mid-prestige positions. We also find that there is less missing data for female cabinet members compared to male cabinet members.

Figure E1 show missingness patterns over time. As noted in the paper, we find that for degree, occupation, and political experience, missingness is decreasing over time. However, we see no such trend in missingness for class.

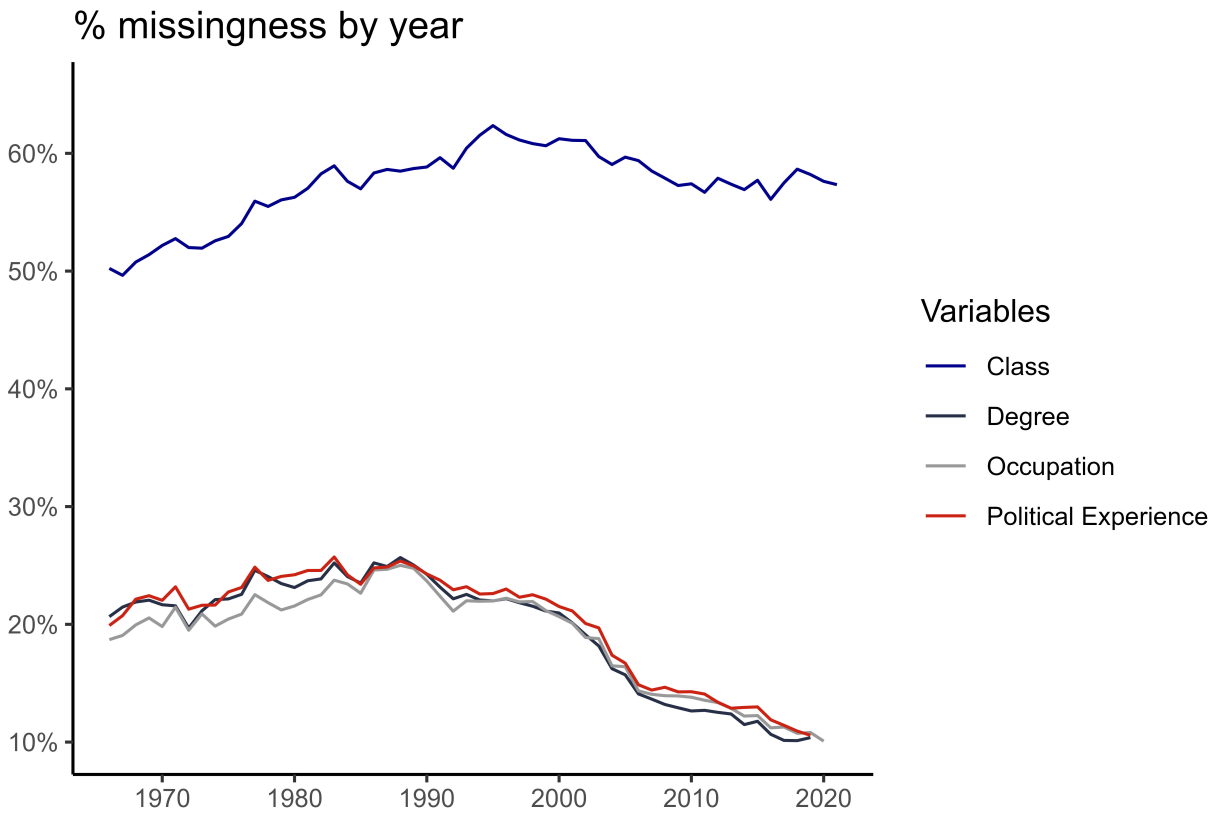


Figure E1: Yearly average missingness for all countries for each variable.

Table E1: Predicting missingness for country-year data

	Education				Class				Occupation			
	(1A)	(1B)	(1C)	(1D)	(2A)	(2B)	(2C)	(2D)	(3A)	(3B)	(3C)	(3D)
Polyarchy VDEM	−0.31*				−0.40*				−0.28*			
	(0.03)				(0.07)				(0.03)			
Log (GDP per capita)		−0.07*				−0.09*				−0.06*		
		(0.01)				(0.01)				(0.01)		
Log(Population)			−0.01				−0.00				−0.01	
			(0.01)				(0.02)				(0.01)	
Africa				0.24*				0.40*				0.23*
				(0.03)				(0.05)				(0.02)
Asia				0.12*				0.19*				0.13*
				(0.03)				(0.05)				(0.02)
Oceania				0.12				0.08				0.09
				(0.10)				(0.15)				(0.10)
North America				0.13*				0.19*				0.14*
				(0.04)				(0.08)				(0.04)
South America				0.13*				0.26*				0.16*
				(0.04)				(0.08)				(0.04)
Constant	0.33*	0.69*	0.41*	0.06*	0.75*	1.24*	0.64*	0.37*	0.31*	0.67*	0.36*	0.06*
	(0.02)	(0.05)	(0.13)	(0.01)	(0.04)	(0.07)	(0.26)	(0.04)	(0.02)	(0.05)	(0.13)	(0.01)
Observations	7114	6696	7035	7144	7114	6696	7035	7144	7114	6696	7035	7144
R2	0.23	0.36	0.01	0.21	0.17	0.26	0.00	0.26	0.19	0.34	0.01	0.20

Reference category for the continents is Europe. Standard errors are clustered by country. *p < 0.05.

Table E2: Predicting missingness for individual-level data

	Education			Class			Occupation			Politician		
	(1A)	(1B)	(1C)	(2A)	(2B)	(2C)	(3A)	(3B)	(3C)	(4A)	(4B)	(4C)
Leader	−0.22*			−0.52*			−0.21*			−0.22*		
	(0.01)			(0.02)			(0.01)			(0.01)		
Female		−0.07*			−0.08*			−0.07*			−0.07*	
		(0.01)			(0.02)			(0.01)			(0.01)	
High prestige			−0.07*			−0.13*			−0.13*			−0.05*
			(0.01)			(0.02)			(0.02)			(0.01)
Medium prestige			0.02			−0.01			−0.01			0.03*
			(0.01)			(0.02)			(0.02)			(0.01)
Constant	0.22*	0.21*	0.20*	0.62*	0.60*	0.66*	0.22*	0.21*	0.66*	0.22*	0.21*	0.19*
	(0.01)	(0.01)	(0.02)	(0.02)	(0.02)	(0.03)	(0.01)	(0.01)	(0.03)	(0.02)	(0.02)	(0.02)
Observations	208801	205638	158992	208801	205638	158992	208801	205638	158992	208801	205638	158992
R2	0.01	0.00	0.01	0.04	0.00	0.01	0.01	0.00	0.01	0.01	0.00	0.01

Reference category for High and Medium Prestige is low-prestige positions. Standard errors are clustered by country. *p < 0.05.

Table E3: Predicting missingness for individual-level data with year-fixed effects

	Education	Occupation	Politician	Class
Female	−0.05*	−0.06*	−0.06*	−0.10*
	(0.01)	(0.01)	(0.01)	(0.02)
Year-fixed	Yes	Yes	Yes	Yes
Observations	205638	205638	205638	205638
R2	0.01	0.01	0.01	0.01

Standard errors are clustered by country. *p < 0.05.

F Extension of Besley and Reynal-Querol (2011)

In this appendix, we replicate the analysis from Besley and Reynal-Querol (2011). In the table below, we assess robustness by using similar econometric specifications as the original paper (Table 1). The first (OLS) specification only controls for log GDP per capita (from Feenstra, Inklaar and Timmer 2015), the second one adds country-specific time trends, and the third one also adds average years of education in the population (from Barro and Lee 2013). Corroborating the original study, our regressions show that democracies have significantly more educated leaders than autocracies. In contrast, we do not find a robust relationship for cabinet ministers.

Moreover, we re-run the analysis changing the dependent variable to whether the leader/minister has attended university at all instead of whether they have obtained a graduate degree. Using these dependent variables, we find that leaders in democracies are significantly more educated than leaders in autocracies, even when holding constant other factors such as income level, education among citizens, and country- and year-fixed effects. In contrast, the gap is smaller when considering the whole cabinet.

Table F1: Regime type and educational attainment: OLS regressions

	Leader			Ministers		
	(1)	(2)	(3)	(4)	(5)	(6)
Democracy	0.18*	0.16*	0.16*	0.06*	0.02	0.01
	(0.04)	(0.05)	(0.05)	(0.02)	(0.02)	(0.02)
Log (GDP per capita)	0.03	0.07	0.05	0.02	0.02	0.02
	(0.05)	(0.08)	(0.08)	(0.02)	(0.02)	(0.02)
Average years of education			0.02*			0.00
			(0.01)			(0.00)
Observations	6080	6080	6049	6112	6112	6081
R^2	0.45	0.56	0.57	0.72	0.79	0.79
Country-specific time trends		Yes	Yes		Yes	Yes

We follow the specifications from Besley and Reynal-Querol (2011). All specifications include country and year dummies and are reported with standard errors clustered at the country level. Standard errors are in parentheses. The dependent variable is either whether the leader has a graduate degree or the share of ministers with a graduate degree. Democracy is a dummy variable that has value 1 if the polity2 score is higher than 0, and 0 otherwise. The full sample is a panel of 141 countries in the period 1966-2018. *p < 0.05.

Table F2: Regime type and educational attainment: OLS regressions

	Leader			Ministers		
	(1)	(2)	(3)	(4)	(5)	(6)
Democracy	0.25*	0.24*	0.24*	0.05*	0.05*	0.05*
	(0.05)	(0.05)	(0.05)	(0.01)	(0.01)	(0.01)
Log (GDP per capita)	0.01	−0.17*	−0.18*	0.01	0.02	0.02
	(0.07)	(0.06)	(0.06)	(0.02)	(0.02)	(0.02)
Average years of education			0.01			0.00
			(0.01)			(0.00)
Observations	6080	6080	6049	6112	6112	6081
R^2	0.45	0.62	0.62	0.56	0.68	0.68
Country-specific time trends		Yes	Yes		Yes	Yes

We follow the specifications from Besley and Reynal-Querol (2011). All specifications include country and year dummies and are reported with standard errors clustered at the country level. Standard errors are in parentheses. The dependent variable is either whether the leader has attended university or the share of ministers who have attended university. Democracy is a dummy variable that has value 1 if the polity2 score is higher than 0, and 0 otherwise. The full sample is a panel of 141 countries in the period 1966-2018. *p < 0.05.

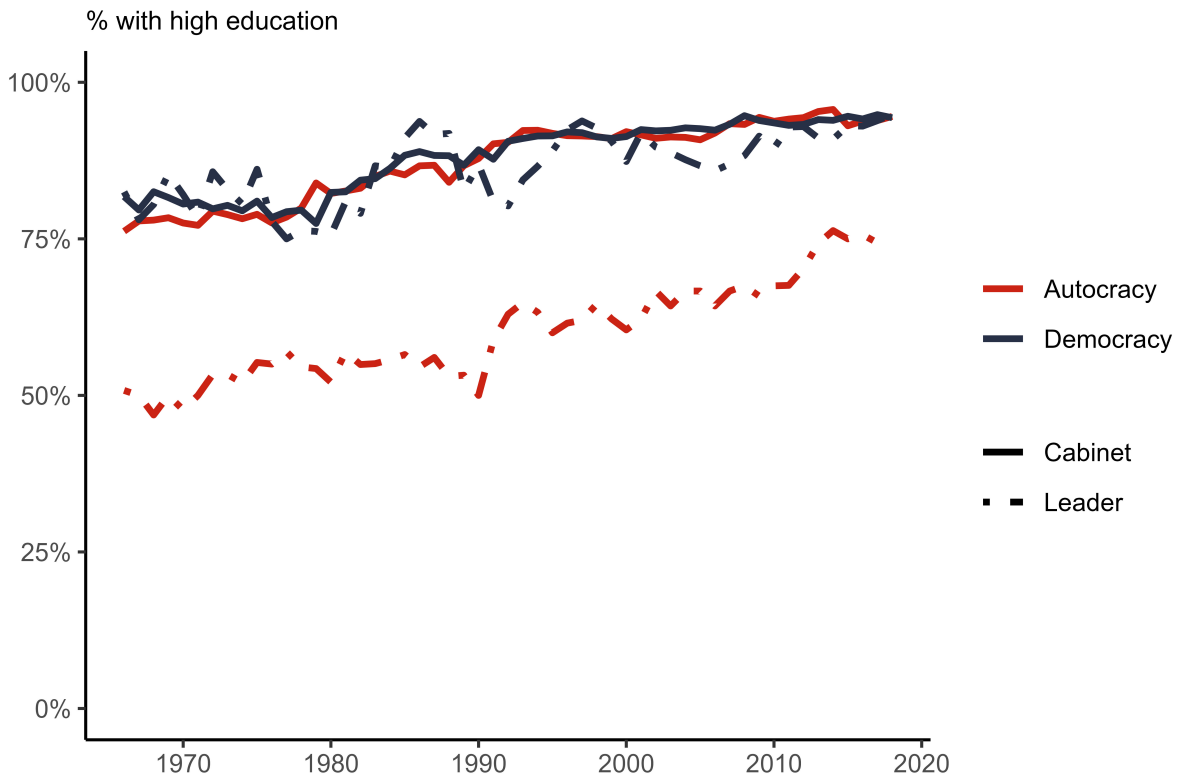


Figure F1: Shares of leaders and cabinet members who attended university, by regime type.

G Inter-coder reliability tests

We conducted an inter-coder reliability test for a stratified (by country), random sample of 400 ministers, randomly selecting and re-coding 10 ministers each from 40 different countries. The countries where chosen through a block-randomization procedure to ensure variation in GDP per capita, population, and democracy levels.

We calculate and present three measures for assessing inter-coder reliability. Firstly, we present the raw percentage agreement between classifications. However, since this measure does not take into account that some agreement may occur due to chance, we also, secondly, calculate Cohen's kappa which accounts for this feature (Cohen 1960), and which is a standard approach for assessing intercoder reliability (Sun 2011). Yet, some of our variables are ordinal rather than nominal, in which case disagreement may be more or less important, depending on the category numbers. This is not something the tests noted above may account for. For instance, the ordinary Cohen's kappa, developed for nominal variables, assumes that it is equally mistaken to code a person with a PhD as having a Master's degree as it would be coding that person as having no education at all. However, a weighted version of Cohen's kappa exists, and this measure weights larger disagreements more heavily than smaller disagreements. Therefore, thirdly, we calculate the weighted kappa whenever applicable (that is, for ordinal variables with more than two categories). All calculations are done with Stata's *kappa* command and weighted kappas are calculated with Stata's prerecorded *w* weights, given by 1 - |i - j|/(k - 1) where i and j index the rows of columns of the ratings by the two raters and k is the maximum number of possible ratings. The kappa equals 0 when the amount of agreement is what is expected to be observed by chance and 1 when there is perfect agreement. We note that we calculated the kappa values with listwise deletion of missing values. This is the standard approach and has been shown in simulation studies to yield only small biases even when data is missing not at random (MNAR), and, importantly, smaller than alternative solutions such as keeping missing values as regular categories when calculating the kappa (see De Raadt et al. 2019).

Table G1 shows the unadjusted percent agreement with Cohen's kappa in parentheses, and where relevant, weighted agreement follows after a slash with the weighted kappa in parentheses. The raw percent agreement is generally high. Many variables, such as *abroad*, *royal*, *political family*, and *sectoral class background* have excellent reliability characteristics with agreements above 90%. The least reliably coded variable is, unsurprisingly, the occupation type variable (60.3%) with 26 categories, some of which are finer grained and some of which constitute broader residual categories. Hierarchical class background and degree score around 70%. But, as displayed in the table, the adjusted agreements are much higher (about 86 and 90%, respectively), indicating that where our coders code differently, they generally do not disagree much (e.g. one coder is coding a minister as having a PhD while the other codes the minister as having a Master's degree).

When we interpret the kappa values, we rely on Landis and Koch (1977). These authors propose that a value of 0 indicates the score that would be expected by chance; poor agreement. 0-0.2 indicates slight agreement; .21-.40 indicates fair agremeent; .41-.60 indicates moderate agreement; .61-.80 indicates substantial agreement; and, .81-1.00 indicates almost perfect agreement.

The lowest observed kappa in the dataset, overall, pertains to hierarchical class background at .55 (although when weighted, this increases to .64), followed by occupation at .55 and degree at .60. These are all challenging variables to code and have only moderate agreement. As discussed in the paper, hierarchical class background, in particular, is difficult to code in a precise manner, since

it typically involves subjective interpretations of biographies or other texts on what constitutes a “working class", “middle class" and “upper class" upbringing. Hence, the moderate agreement is to be anticipated, and users of the dataset should take care to account for, or at least acknowledge, measurement errors (and its implications, such as potential attenuation bias when used as a covariate in regression analysis).

The remaining variables range from displaying substantial agreement (political experience, degree type, and sectoral class background) to almost perfect agreement (political family, abroad, royal). A few kappas are estimated to be exactly 0. Note that this is not an error. The extreme distribution of some variables, such as royal background in low income countries (100 no, 2 yes) means that expected agreement is as high as the real agreement (98%), rendering a kappa value of 0.

We then evaluate whether intercoder reliability is different for different types of countries. On average, reliability is higher in richer, more populous, and more democratic countries,. Yet we also note that these differences across countries are rather modest and also not consistent: For instance, reliability is higher for the variable *degree* in less wealthy, less populous, and less democratic countries. Users of the dataset should thus consider the patterns for their particular variables of interest when discussing the extent to which reliability characteristics in the data are systematic (and, in what manner) and considering potential implications for inferences.

Table G1: Inter-coder reliability test. Raw percent agreement (unadjusted Cohen's kappa) / weighted agreement (adjusted kappa)

Variable	Overall	High inc. countries	Low inc.	High pop.	Low pop.	High Dem.	Low Dem.
Degree	67.4% (.60) / 90.2% (.57)	64.5% (.56) / 88.7% (.50)	72.5% (.66) / 92.9% (.69)	62.3% (.53) / 89.1% (.53)	73.4% (.67) / 91.5% (.62)	61.4% (.52) / 87.8% (.45)	74.4% (.67) / 92.5% (.70)
Degree type	73.6% (.70)	77.6% (.74)	66.7% (.62)	80.7% (.78)	64.9% (.60)	74.8% (.70)	72.2% (.69)
Abroad	92.4% (.85)	94.7% (.89)	87.5% (.74)	96.9% (.94)	86.6% (.73)	93.6% (.85)	91.0% (.81)
Royal	99.0% (.88)	99.4% (.96)	98.1% (.00)	98.8% (.74)	99.2% (.94)	98.7% (.00)	99.3% (.96)
Political family	95.3% (.81)	96.0% (.86)	93.7% (.47)	93.9% (.74)	96.9% (.89)	95.0% (.70)	95.7% (.87)
Class background, hierarchical	71.1% (.55) / 85.6% (.64)	72.6% (.57) / 86.3% (.66)	64.7% (.42) / 82.3% (.51)	65.3% (.45) / 82.7% (.53)	78.0% (.67) / 89.0% (.75)	67.9% (.49) / 84.0% (.57)	75.7% (.56) / 87.8% (.66)
Class background, sector	95.6% (.78)	94.5% (.64)	100% (1.0)	93.9% (.63)	97.6% (.90)	96.2% (.81)	94.6% (.72)
Occupation	60.3% (.55)	64.0% (.60)	54.1% (.46)	64.3% (.60)	55.7% (.47)	60.4% (.56)	60.2% (.52)
Political exp.	79.1% (.65)	80.5% (.65)	76.7% (.63)	84.4% (.69)	72.7% (.58)	80.3% (.61)	77.7% (.64)

References for the appendix

Barro, Robert J and Jong Wha Lee. 2013. "A new data set of educational attainment in the world, 1950–2010." *Journal of Development Economics* 104:184–198.

Besley, Timothy and Marta Reynal-Querol. 2011. "Do democracies select more educated leaders?" *American Political Science Review* 105(3):552–566.

Cohen, Jacob. 1960. "A coefficient of agreement for nominal scales." *Educational and psychological measurement* 20(1):37–46.

De Raadt, Alexandra, Matthijs J Warrens, Roel J Bosker and Henk AL Kiers. 2019. "Kappa coefficients for missing data." *Educational and Psychological Measurement* 79(3):558–576.

Feenstra, Robert C., Robert Inklaar and Marcel P. Timmer. 2015. "The Next Generation of the Penn World Table." *American Economic Review* 105(10):3150–3182.
URL: *www.ggdc.net/pwt*

Gerring, John, Erzen Oncel, Kevin Morrison and Daniel Pemstein. 2019. "Who rules the world? A portrait of the global leadership class." *Perspectives on Politics* 17(4):1079–1097.

Landis, J Richard and Gary G Koch. 1977. "The measurement of observer agreement for categorical data." *Biometrics* pp. 159–174.

Nyrup, Jacob and Stuart Bramwell. 2020. "Who governs? a new global dataset on members of cabinets." *American Political Science Review* 114(4):1366–1374.

Sun, Shuyan. 2011. "Meta-analysis of Cohen's kappa." *Health Services and Outcomes Research Methodology* 11:145–163.

WHED. 2024. "World Higher Education Database.". 19/02/2024.
URL: *https://www.whed.net/home.php*